

Putting knowledge and technology to work: A look at EU innovation output

HIGHLIGHTS

- Tracking innovation performance is high on the European Commission's agenda.
- According to the Innovation Output Indicator, developed by the European Commission since 2013, the EU as a single economic block leads New Zealand, China, Australia and Brazil in 2021, but lags behind its main competitors, including the United States, South Korea and Japan.
- The EU economy performs well regarding the domestic contribution to the value of medium-high and high-tech manufacturing exports, but should improve the share of employment in knowledge-intensive industries as well as the percentage of innovation-active firms.
- The innovation divide persists among EU countries. Sweden, Germany and Finland are the best performers, while Romania, Latvia and Bulgaria are the EU Member States reporting the lowest performance in 2021.

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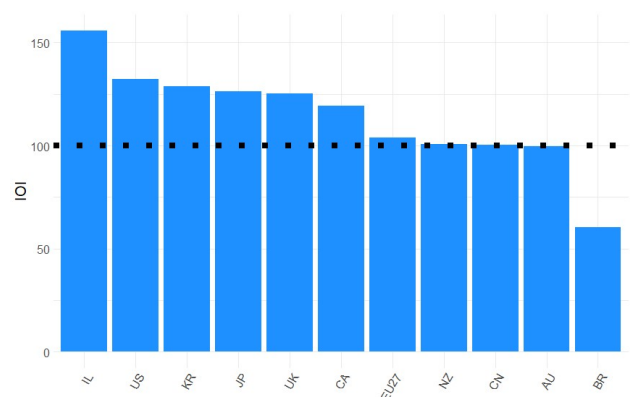
EU Innovation Output

Innovation plays an important role in tackling the most pressing social, economic and environmental challenges. This is why tracking innovation performance is high on the European Commission's agenda. Since 2013, the European Commission produces the Innovation Output Indicator (IOI) - a composite indicator offering an output-oriented measure of innovation performance at the country and EU levels.

The IOI tracks the extent to which new ideas reach the market, providing better jobs and making Europe more competitive. It aggregates four components including 1. patents and trademarks, 2. employment in knowledge-intensive activities, 3. the domestic contribution to the value of knowledge-intensive exports, and 4. Innovation-active enterprises.

Each year the IOI is used to compare the innovation output of the EU as a block to other world players. As depicted in Figure 1, the EU leads New Zealand, China, Australia and Brazil in terms of innovation output in 2021, but still lags behind its main competitors, including the United States, South Korea and Japan.

Figure 1 – Innovation Output Indicator of the EU in 2021, compared to international competitors



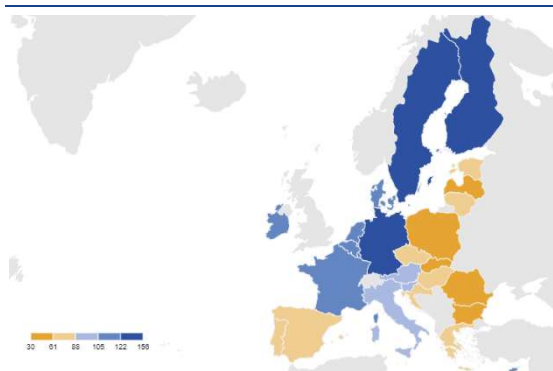
Source: Bello et al. 2023. Note: dotted line shows the EU27 value in 2011.

QUICK GUIDE – Composite indicators and scoreboards are widely used to capture both the process and the outcomes of complex phenomena such as innovation. Since 2013, the Joint Research Centre of the European Commission has produced the Innovation Output Indicator within the INNOVA projects to offer an output-oriented measure of innovation for the EU and its Member States. The IOI measures the extent to which ideas for new products and services are able to reach the market, provide better jobs and make Europe more competitive.

In its latest edition, the IOI includes six indicators combined in four components: 1) the number of Patent Cooperation Treaty (PCT) patent applications and the number of trademark classes in total resident trademark applications, each denominated by population (in millions); 2) the number of persons employed in knowledge-intensive activities and business industries over total employment; 3) domestic value added content of medium-high and high-tech manufacturing exports as a share of total manufacturing exports and domestic value added content of knowledge-intensive service exports as a share of total service exports; and 4) the number of innovation-active firms over the total number of firms.

Among EU Member States (MSs), Sweden, Germany and Finland are the best performers (Figure 2). Patent applications drove the top performance of Sweden, partly due to its exceptionally strong innovation in digital communication [7]. The domestic contribution to the value of knowledge-intensive manufacturing exports drove Germany's high performance, reflecting its strong industrial base in basic metals, chemicals and motor vehicles [6]. Finland, on the other hand, is very strong in terms of its share of innovation-active enterprises in the economy, reflecting its government's commitment to fostering an innovation ecosystem [5]. Conversely, Romania, Latvia, Poland and Bulgaria are the EU MSs reporting the lowest performance in 2021.

Figure 2 – Innovation Output Indicator of EU Member States, 2021



Source: The JRC Composite Indicators & Scoreboards Explorer, <https://ec.europa.eu/scoreboards-explorer>, accessed on 26th June 2023

A closer look at the IOI components

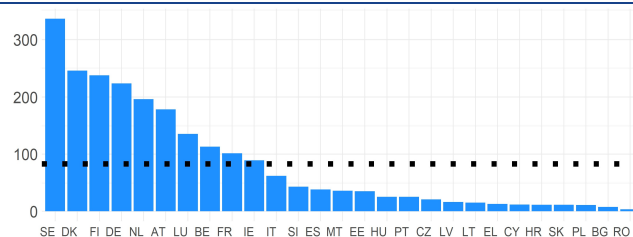
EU MSs show a wide range of innovation performance in the IOI, as well as in its components.

IP applications

According to 2021 IOI results, Sweden is leading the ranking in terms of number of patent applications per million population, while a large number of countries exhibits a particularly low performance, with Romania and Bulgaria being at the bottom of the distribution with nearly no Patent

Cooperation Treaty (PCT) patent applications (Figure 3). By contrast, the innovation divide among MSs is less pronounced in trademark applications with Luxembourg leading the ranking, followed by Malta [2].

Figure 3 – Number of patent applications filed under the Patent Cooperation Treaty per million population, 2019



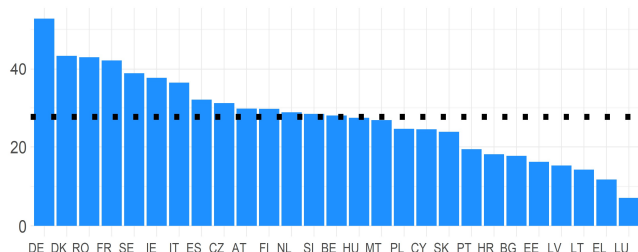
Source: Bello et al. 2023. Note: The dotted line shows the EU27 average.

Contribution to knowledge-intensive exports

In 2021, around half of total EU product exports and three-quarters of EU service exports were knowledge-based [1]. The EU MSs, however, still show a high intra- and extra- EU dependence in the production and exports of these products and services. One example is that the domestic contribution to the value of medium-high and high-tech manufacturing exports in EU countries is less than one-third on average. Germany is the EU country that depends the least on foreign inputs. But when looking at the EU as a block, its performance is comparable to the USA's as over half of the value of knowledge-intensive manufacturing exports comes from within its own economy.

When it comes to the service sector, the EU MSs' domestic contribution to the value of knowledge-intensive service exports is slightly higher. Finland is least dependent, with over two fifths of their value being domestic. In comparison to other countries like China, Japan, and Canada, the EU as a whole is slightly ahead with a domestic value contribution close to 40%. However, the EU falls behind Israel, the USA, and the UK where over half of the value of knowledge-intensive service exports is domestically contributed.

Figure 3 – Domestic value added content of exports in high and medium-high technology manufacturing (MHT) industries as percentage of total manufacturing exports, 2018



Source: Bello et al. 2023. . Note: The dotted line shows the EU27 average.

Employment in knowledge-intensive activities

Employment in knowledge-intensive activities and industries is an additional challenge for EU MSs, as they make up less than 15% of total employment in the EU on average. This is well below other countries like Israel (34%), South Korea (22%) and the United States (17%). The top performers among EU MSs are Luxembourg (26%) and Sweden (21%).

Innovation-active enterprises

Innovation can happen in all sectors, including those that are less knowledge-intensive. The share of innovative enterprises aims to capture product and business process innovation activities across the entire economy and is the newest additional indicator in the IOI. According to 2021 IOI results, nearly half of all firms in the EU have been identified as innovation-active, placing the EU ahead of some international competitors such as South Korea, Brazil, and China, and at par

Box 1: Impact of Covid

The COVID-19 pandemic crisis has largely disrupted the business environment, as well as the functioning of the science and innovation systems. R&D investment (PPS, in constant prices), which tends to be pro-cyclical, declined by 2% in 2020, whereas the share of R&D expenditure on GDP increased by nearly 3.6%, mainly due to a stronger drop in GDP (by nearly 4%).

Preliminary data also seems to point to a decline in the competitiveness of EU knowledge-intensive exports. The share of medium and high-tech product exports declined by 3.4% in 2021, which is the highest drop in the last 10 years. Similarly, the share of these sectors in total employment declined by nearly 2%. Finally, the crisis also seems to have affected business dynamics in the EU with the employment share in newly-established firms and that of fast-growing firms declining by 15% and 19%, respectively, in 2020.

The effects of the pandemic crisis on EU countries' innovation performance is not yet captured in the IOI as only two of six indicators, refer to the pandemic period.

with current world leader Israel, but still falling behind the USA and Canada. However, as a block, the EU's performance hasn't shown much improvement, with the latest data (from 2018) only 2 percentage points higher than in 2011 [2]. There are significant variations among EU MSs, with Greece and Belgium having the highest shares at around 60% - while Romania has the lowest performance at only 14%.

Conclusions

The Innovation Output Indicator tracks the extent to which new ideas reach the market, providing better jobs and making Europe more competitive. According to the 2021 IOI, Finland, Sweden and Germany are clear leaders within the EU while Romania, Latvia and Bulgaria have the largest gap to close.

When looking at the individual components of IOI, EU MSs have their unique strengths and weaknesses. On some components, EU MSs are among the best performers globally: of the top ten countries with the highest percentage of innovation-active enterprises, six are EU MSs; Sweden is a world top performer in terms of patents per million population; and Germany competes with the US in terms of the domestic contribution to the value of its medium-high and high-tech manufacturing exports.

Some individual EU MS weaknesses are alleviated by the overall strength of the Union. In particular, the EU as a single block does well in its domestic contribution to the value of high-tech manufacturing exports. However, to be a world leader in innovation output, it must close its disparities with the top performers on other components of the IOI and improve the share of employment in knowledge-intensive industries as well as the percentage of innovation-active firms.

Related work

This work is a deliverable of the INNOVA MEASURE V project (Horizon Europe 101052304). More information can be found on: https://knowledge4policy.ec.europa.eu/projects-activities/innova-project_en

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